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RESEARCH ARTICLE

LASSO-mCGA: Machine Learning and Modified Compact Genetic Algorithm-Based Biomarker Selection for Breast Cancer Subtype Classification

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ABSTRACT Breast cancer is the most common cancer type among females and is one of the leading causes of death worldwide. Being a heterogeneous disease, subtyping breast cancer plays a vital role in its treatment. In this regard, gene expression plays an important role. Thus, in this work gene expression data is used to identify the most significant gene biomarkers. The identified biomarkers are highly associated with each breast cancer subtype such as Luminal A, Luminal B, HER2-Enriched and Basal-Like. To identify such biomarkers, initially LASSO in association with four machine learning models such as Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbours (KNN) and Naive Bayes (NB) are applied on the dataset to find the initial reduced set of genes as well as the best learning model based on classification accuracy; SVM in this case. Thereafter, Modified Compact Genetic Algorithm (mCGA) is performed to identify the final set of genes as biomarkers for each specific subtype. Experimental results suggest that our proposed method assesses AUC-ROC values of 0.9878 and 0.97311 for LumA and LumB and 1 for Basal and HER2 subtypes. To validate the biological significance of the identified biomarkers, KEGG pathway and GO enrichment analysis are carried out.

INDEX TERMS Biomarker, breast cancer, feature selection, LASSO, modified compact genetic algorithm.

I. INTRODUCTION

Cancer research has emerged as one of the most important areas of study in medicine over the last few decades [1]. Generally speaking, cancer is defined biologically as an abnormal cell growth caused by cell divisions in any part of the body [2]. While cell development can be relatively slower in some types of cancer, it can be extremely quick in others. Each cell in the human body has a particular function as well as a lifespan. Apoptosis is a natural phenomenon by which a cell dies after a specific amount of time [3]. Any unnatural genetic change, however, can impair apoptosis and upset the balance between cell division and cell death [3], [4]. This may lead to an uncontrolled growth as tumour. Additionally, inherited genetic flaws and hormonal problems are important contributors to the emergence of cancer. Breast cancer is the most common cancer type among females [5]. It has surpassed lung cancer in 2020 and represents 11.7% of all cancer cases [6]. Studies have concluded that the global burden of breast cancer is expected to cross 2 million by 2030 [7]. As per data, breast cancer accounted for 13.5% of all cancer cases and 10.6% of all deaths [5]. Breast cancer turns out to be fatal when it metastasises and spreads to other parts of the body. Based on the molecular receptors, breast cancer can be further divided into four subtypes [8]: Luminal A, Luminal B, HER2-Enriched and Basal-Like. Luminal A (LumA) has slow progression and it usually is Estrogen Receptor (ER) positive, Progesterone Receptor (PR) positive and Human Epidermal growth factor Receptor 2 (HER2) negative while Luminal B (LumB) has

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a faster growth than LumA and is ER positive, PR negative and HER2 positive. HER2-Enriched (HER2-E) subtype is HER2 positive but ER and PR negative. For, Basal-Like (BL), ER, PR and HER2 are all negative. Thus, it is also called triple negative breast cancer. Classifying cancer subtypes can provide an accurate and efficient diagnosis for different types of cancer [9], [10]. As the biologic subtype of cancer contributes significant prognostic and predictive information for patients with breast cancer [11], classifying breast cancer into its subtypes and identifying key genes associated with it are essential for the correct and efficient treatment of patients. In this regard, gene expression data is a viable option for cancer classification. However, gene expression data consists of complex and high-dimensional matrices of genes versus samples. The number of samples is typically substantially lower than the number of genes [12]. Cancer classification using gene expression data is usually either binary (cancer/normal) [13], [14], [15] or multiclass (subtypes) [15], [16] classification. The samples of expression data may have identical histopathological appearances but very diverse clinical outcomes, making multiclass problem more difficult to solve [17]. Along with cancer classification task, feature selection methods are also widely used in the literature.

Feature selection is an important technique in machine learning and data analysis that selects a subset of relevant features from an original set of variables. The most informative and discernible features are selected to improve the performance of the predictive model [18], [19]. Feature selection also reduces overfitting of machine learning models, reducing the dimensionality of the dataset [20]. Recent studies have used genetic algorithms to optimize feature set by reducing the dimensionality of datasets and improving model accuracy. Genetic algorithms (GA) can be used to improve traditional machine learning models by integrating feature selection mechanisms [21], [22]. This process involves using GA to find the best subset of features that maximizes model performance. Recently, Dey and Mukhopadhyay [23] have proposed a compact genetic algorithm (CGA) based on learning vector quantization (LVQ) that selects a subset of features that improves the accuracy of traditional machine learning models, especially in weighted random forests. Boruta algorithm has been used in many studies to select important features [24]. For high-dimensional data, the L1-norm SVM is a particularly effective feature selection algorithm. It can provide highly related features with very high accuracy [25], [26].

In [27], Graudenzi et al. have used Support Vector Machines and feature selection based on information about relevant pathways in breast cancer to identify subtypes. In this regard, they have identified 400 genes from feature selection method and achieved an overall accuracy of 85%. Feature selection along with attention mechanism have been used by Lee et al. [28] to classify five cancer types. The accuracies for such classification for different cancer types are 66.91% for breast invasive carcinoma (BRCA), 76.39% for urothelial bladder Carcinoma (BLCA), 78.13% for (Stomach adenocarcinoma) STAD 70.12% for prostate adenocarcinoma (PRAD) and 71.54% for colon adenocarcinoma (COAD). Mostavi et al. [15] have used 1D CNN to identify breast cancer subtypes. They have used feature selection based on statistical methods like mean and standard deviation which resulted in 7091 genes. The average accuracy achieved by such model is 88.42% among five subtypes. In their work, Shukla et al. [29] have initially employed ensemble gene selection (EGS) method on colon cancer dataset using multilayer approach and F1 score to filter redundant genes. Thereafter, an adaptive genetic algorithm (AGA) is used to identify final set of genes. They have employed SVM and achieved an accuracy of 89%. Pareto-front analysis have been used by [16] for gene selection and thereafter SVM is used for classification of multi-tissue, lung and leukemia datasets. The model have achieved an F1 score of 95.11 for 397 genes in the mixed lineage leukemia dataset. A hybrid $L_{1/2+2}$ regularisation (HLR) function have been used by Huang et al. [30] to select set of genes to apply logistic regression. In this regard, three different binary classification datasets have been used and the method achieved more than 89% accuracy in all the tests. Outlier based gene selection (OGS) method along with SVM have been used in [31] for breast cancer subtype classification. The F1 score achieved by such method is 1 for Basal and 0.86 for HER2.

Motivated by the literature, in this work we have employed LASSO based Modified Compact Genetic Algorithm (LASSO-mCGA) for feature/gene subset selection and thereafter used SVM to classify the breast cancer subtypes and identify the final set of genes as biomarkers. In this regard, initially LASSO is used to identify the initial reduced set of genes for each of the breast cancer subtypes, LumA, LumB, HER2 and Basal. Thereafter, four machine learning models, RF, SVM, KNN and NB are applied on the reduced gene set to identify the best model where SVM performs better in terms of Accuracy, Precision, Recall and F1 score. Subsequently, mCGA along with SVM are applied on the reduced set of genes as identified by LASSO to find the final set of genes as biomarkers for each specific subtype. Experimental results show that the proposed method achieves AUC-ROC values of 0.9878 and 0.9731 for LumA and LumB while AUC-ROC of 1 is achieved for both HER2 and Basal. Finally, KEGG pathways and GO enrichment analysis are also carried out in order to validate the biological significance of the identified genes as biomarkers.

II. MATERIALS AND METHODS

In this section, the preparation of gene expression dataset is put forth along with the discussion of the complete pipeline of this work.

A. DATASETS

To conduct the experiments in this work, Breast Invasive Carcinoma (BRCA) dataset is considered. The dataset contains four molecular subtypes of breast cancer viz.

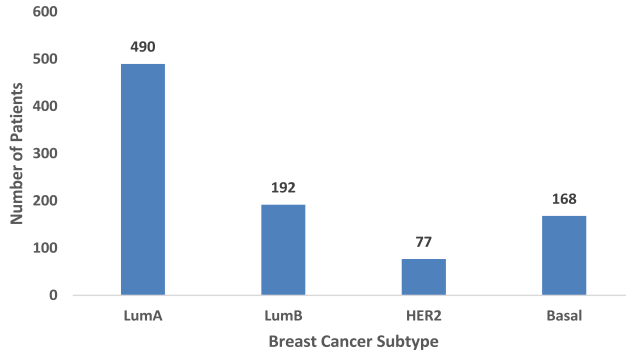


FIGURE 1. Statistics of BRCA dataset.

Luminal A (LumA), Luminal B (LumB), HER2-Enriched (HER2) and Basal-Like (Basal). The dataset also includes data of Normal patients who are not affected by breast cancer. However, as this work focuses on breast cancer subtype classification, the data of Normal patients are not considered. The Next Generation Sequencing (NGS) based gene expression values and clinical data are compiled from The Cancer Genome Atlas (TCGA).[1] The common patients between gene expression and clinical datasets are considered for further processing. The data as collected from TCGA have genes with no expression values. Thus, the genes with more than 75% non expressed values are removed from the dataset. This resulted in 28201 genes with 927 subtype expression values. Figure 1 reports the number of breast cancer patients (samples) per subtype. The codes and final list of biomarkers are available as Supplementary at https://github.com/NimishaGhosh/LASSO-mCGA. The data files are available on request.

B. PIPELINE OF THE WORK

The pipeline of the work is given in Figure 2. The primary motivation of this work is to correctly identify biomarkers for each subtype which can accurately predict such breast cancer subtypes. In this regard, initially feature selection using Least Absolute Shrinkage and Selection Operator (LASSO) is performed. Thereafter, Modified Compact Genetic Algorithm (mCGA) is considered to identify the final set of gene biomarkers. Once the final set of biomarkers are identified, their biological significance is verified using KEGG pathway and GO enrichment analysis. The feature selection methods as employed in this work are explained next.

1) FEATURE SELECTION USING LASSO-mCGA

Selection of optimal features to accurately identify breast cancer subtype and thereby finding the biomarkers is quite a challenging task. The number of features in this work corresponds to the number of genes. Since the number of features is huge (28201), finding the optimal subset

[1]https://cancergenome.nih.gov/

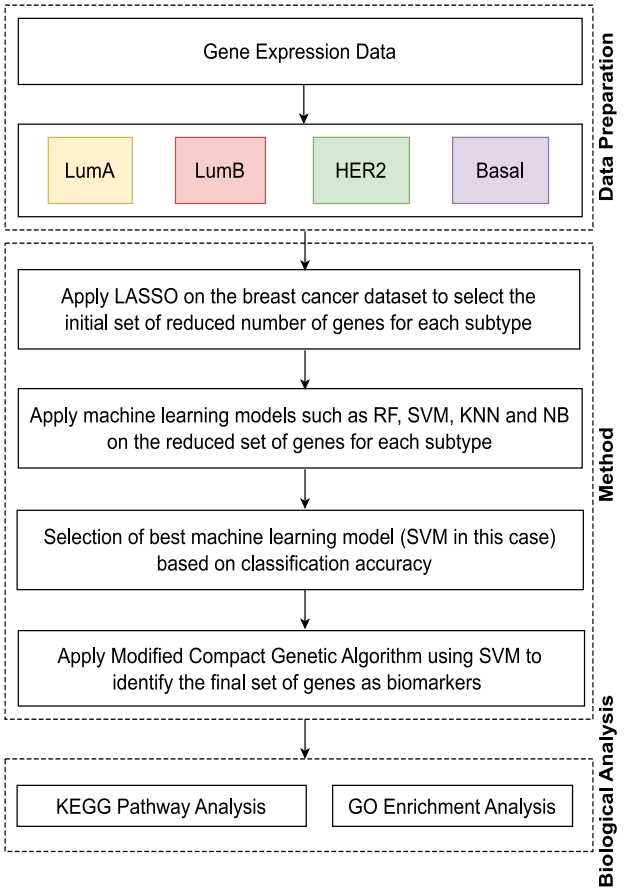


FIGURE 2. Pipeline of the work.

of features is an NP-Hard problem. Numerous methods have been used in the past to address this problem. However, due to their high time and space requirements, exhaustive search or brute force strategies are ineffective for high-dimensional data while genetic algorithms (GA) and particle swarm optimisation (PSO) have shown to achieve near-optimal accuracy for such data [32]. Although GA has shown promising results, it stores the whole population in each iteration thereby having a high memory requirement. PSO also suffers from the same shortcomings. To overcome these issues, Harik et al. [33] have proposed Compact GA (CGA), which rather than storing the entire population, saves a probability vector representing the same. CGA requires less memory than ordinary GA because it depicts the prevalence of each gene in the population. However, the straightforward implementation of CGA may have lack of exploration and thereby CGA may be trapped in local maxima. To mitigate these issues, in this work Modified Compact Genetic Algorithm (mCGA) [34] is used. The main purpose of mCGA is to improve the exploratory ability of CGA.

It is to be noted that as the number of features is very large, even mCGA may take a very long time to converge. Please also note that feature selection techniques

like Recursive Feature Elimination (RFE) and Learning Vector Quantization (LVQ) may take days to provide results on very huge datasets with large number of features. Thus, to further minimise the time complexity of mCGA and choose the appropriate biomarkers, initially we apply LASSO to the dataset to find the initial set of reduced features. The main idea behind LASSO is to minimise a cost function as given in Equation 1 [35], thereby retaining only those features which are useful and categorically remove the redundant ones. However, to appropriately use LASSO, the hyperparameter α has to be properly tuned. This is explained in the Results Section. Thereafter, all the four models RF, SVM, KNN and NB are applied on the reduced set to find the best machine learning (ML) model based on classification accuracy; SVM in this case. All the experiments are performed by using 10 fold cross validation repeated 10 times.

$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2+\alpha\sum_{j=1}^{p}|\beta_j| \tag{1}$$

Here, n is the number of samples in the dataset, y_i is the target variable, β_0 and β_j are the bias and coefficients respectively, p is the number of features while x_{ij} represents the feature vector.

After applying LASSO and finding the suitable ML model along with the initial set of features/genes; 97 for LumA, 80 for LumB, 126 for HER2 and 188 for BASAL, mCGA is applied on the same to identify the final set of biomarkers. mCGA is an extension of CGA where in the first step four probability vectors (PVs) of size $\mathcal{N}$ are initialised with a value of 0.5, where $\mathcal{N}$ is the total number of features. In the second step, based on the four PVs, four binary strings (where 1 and 0 indicate presence and absence of a feature) of size $\mathcal{N}$ are randomly generated. In the third step, the four PVs are compared with each other based on a fitness function. In this work, fitness function is calculated using value of the Area Under the Curve of the Receiver Operating Characteristic (AUC-ROC), which is a plot of the false positive rate versus the true positive rate for various classification probability threshold values. The AUC-ROC value is determined by employing SVM where the value lies between 0 and 1. The fourth step consists of updating each of the four PVs based on the comparison. This step considers updating each element of each PV such that the update takes place both in accordance with the standard cGA's rule (reflecting its self-knowledge) and elements of the other PVs (reflecting global knowledge) as well, particularly with the best element among those of all the PVs. This ensures a local as well as a global update. Here, the basic idea of Particle Swarm Optimisation is applied where the global update is particularly dependent on the learning factor c which is taken as 2 in accordance with [34]. Starting from Step 2, these procedures are repeated till any of the PVs converge. The converged PV represents the final set of biomarkers. Thus using mCGA, it is possible to increase the exploration qualities of the algorithm as well as enhance its ability to avoid local optimum while having a nominal increase in computing cost as mentioned in [34]. The algorithm for selecting the biomarkers using LASSO-mCGA is provided in Algorithms 1 and 2.

Algorithm 1 Selecting Biomarkers for Breast Cancer Subtypes Using LASSO-mCGA

Input: The set of genes as features
Output: Reduced subset of genes, Best ML model

1. Find the best hyperparameter α for LASSO.
2. Apply LASSO to find the initial reduced subset of genes.
3. Apply ML models to identify the best one based on classification accuracy.
4. Apply mCGA (Algorithm 2) on the selected subset of genes to identify the final set of biomarkers.

Algorithm 2 Algorithm for mCGA to Find the Final Set of Biomarkers

Input: Reduced subset of genes from Algorithm 1
Output: Final set of biomarkers

1. Initialise 4 probability vectors of length $\mathcal{N}$ with 0.5.
2. Randomly generate 4 binary strings of length $\mathcal{N}$ and calculate their fitness values.
3. Compare the PVs based on the fitness values; winner: PV with the corresponding binary string having the best fitness, loser: the other 3 PVs.
4. PV Update:

```
Local update
for i ∈ 1…PV do
  for j ∈ 1…𝒩 do
    if winner(i,j) ≠ loser(i,j) then
      if winner(i,j) == 1 then
        PV(i,j) = PV(i,j) + 1/n
      else
        PV(i,j) = PV(i,j) - 1/n
      end if
    end if
  end for
end forGlobal update
Best_PV = winner
for i ∈ 1…PV  do
  PV(i) = PV(i) + c(Best_PV - PV(i))
end for
```

5. Repeat steps 2 to 4 till any of the PV converges.
6. The converged PV provides the final set of biomarkers.

2) PERFORMANCE METRICS

The different performance parameters as used in this work are Accuracy, Precision, Recall and F1 score. These parameters can be evaluated as:

$$Accuracy=\frac{TP+TN}{TP+FP+TN+FN}\times 100\% \tag{2}$$

Accuracy provides the ratio of correctly identified subtypes to all the evaluated subtypes.

$$Precision = \frac{TP}{TP + FP} \times 100\% \tag{3}$$

Precision is the ratio of correctly predicted breast cancer subtypes to the total number of predicted breast cancer subtypes.

$$Recall = \frac{TP}{TP + FN} \times 100\% \tag{4}$$

Recall is the ratio of correctly predicted breast cancer subtypes to all of the breast cancer subtypes.

$$F1Score = \frac{2 * (Recall * Precision)}{Recall + Precision} \tag{5}$$

F1 Score is the weighted average of Precision and Recall.

III. RESULTS AND DISCUSSION

This section summarises and discusses the different results of this work.

A. FEATURE SUBSET SELECTION USING LASSO-mCGA

The initial results while considering all the features and all the subtypes taken together are reported in Table 1 while Table 2 reports the results for each of the subtype after executing LASSO and the four machine learning algorithms. Table 2 also reports the hyperparameter α for each of the subtype.

TABLE 1. Comparison of performance between the four machine learning models considering all the features.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score
RF	88.70	88.94	88.70	0.8790
SVM	**90.32**	**90.14**	**90.32**	**0.9005**
KNN	84.94	84.87	84.94	0.8410
NB	81.18	84.48	81.18	0.8190

As can be seen from Table 1, SVM with Accuracy, Precision, Recall and F1 Score values of 90.32, 90.14, 90.32 and 0.9005 respectively shows the best performance amongst all. It is to be noted that different hyperparameters like number of trees for RF, regularisation parameter for SVM, K for KNN, and NB parameters are deduced experimentally; linear kernel is used for SVM with the regularisation parameter C as 7. Statistical significance is also tested to analyse the results as obtained from the four classifiers. In this regard, Wilcoxon rank-sum test which is a non-parametric test for significance assessment is used in this work. The pairwise p-values generated by such a test between SVM and RF, KNN and NB are 1.28E-13, 3.72E-18 and 3.77E-18 respectively. It can be observed that all pairwise p-values are less than 0.05, thereby leading to the conclusion that median of these distributions differ and these are not random occurrences.

It can be seen from Table 2 that α values for LASSO as found experimentally by LASSOCV pipeline of Python are 0.044, 0.133, 0.039 and 0.010 for LumA, LumB, HER2 and Basal respectively. Similarly, by considering such values, SVM shows the best performance here as well in classifying all the subtypes. Thus, SVM is considered as the learning model while performing mCGA. For SVM, linear kernel is considered with the regularisation parameter C as 2 for LUMA, 8 for LUMB and 6 for both HER2 and Basal; all of these values are found experimentally.

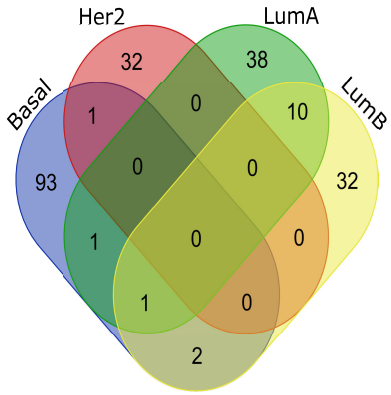


FIGURE 3. Venn diagram to show the common biomarkers among the four subtypes.

The results after applying SVM on the final set of genes/features as obtained by mCGA are reported in Table 3. Table 3 also reports the results as obtained by applying SVM on PAM50 genes [11]. The number of genes/features as deduced by mCGA for LumA, LumB, HER2 and Basal are 50, 45, 33 and 98 respectively. These will hereafter be called as biomarkers. The full list of biomarkers and the list of PAM50 genes are provided in the Supplementary. It can be observed from the Table 3 that with a classification accuracy of 100%, SVM can correctly identify all HER2 and Basal subtypes based on these biomarkers. Even with luminal types which are very similar to each other, the proposed approach shows promising results with an accuracy of 98.92% for LumA and 97.85% for LumB. The values of other metrics like Precision, Recall, F1 score and AUC-ROC for LumA are 98.95%, 98.92%, 0.9892 and 0.9878 respectively while for LumB the values for the same metrics are 97.99%, 97.89%, 0.9785 and 0.9731 respectively. On the other hand, although PAM50 genes can correctly identify Basal subtypes with 100% accuracy, for the other subtypes, the proposed approach performs competitively better. This further goes to show that a common set of genes fails to accurately identify different breast cancer subtypes. Figure 3 shows the common biomarkers among the four subtypes. As can be seen from the figure, LumA and LumB share 10 common biomarkers between them. On the other hand, HER2 and Basal share only 1 common biomarker (KRT20), so does LumA and Basal (TPSD1) and LumA, LumB and Basal (NTS) while there are 2 common biomarkers between LumB and Basal (RASSF10, FOXJ1). It is worth mentioning here that there are 3 common genes between LumA and PAM50 genes (CDC6, KRT17, PGR) and LumB and PAM50 genes (SFRP1, BIRC5, KRT5)

TABLE 2. Comparison of performance between the four machine learning models after applying LASSO.

Subtype	α	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score
LumA	0.044	RF	94.11	94.24	94.11	0.9410
		SVM	**95.63**	**95.73**	**95.63**	**0.9562**
		KNN	91.05	91.34	91.05	0.9101
		NB	94.36	94.51	94.36	0.9435
LumB	0.133	RF	92.50	92.61	92.50	0.9202
		SVM	**94.85**	**94.95**	**94.85**	**0.9483**
		KNN	90.89	90.80	90.89	0.9033
		NB	91.16	92.80	91.16	0.9156
HER2	0.039	RF	96.95	96.98	96.95	0.9665
		SVM	**98.77**	**98.83**	**98.77**	**0.9875**
		KNN	97.58	97.58	97.58	0.9745
		NB	98.26	98.55	98.26	0.9833
Basal	0.010	RF	99.66	99.67	99.66	0.9966
		SVM	**99.76**	**99.76**	**99.76**	**0.9975**
		KNN	99.66	99.67	99.66	0.9966
		NB	99.73	99.73	99.73	0.9972

TABLE 3. Classification results of the four subtypes using SVM on PAM50 genes and the genes obtained by mCGA.

	Accuracy (%)				Precision (%)				Recall (%)				F1 Score				AUC-ROC			
Method	LumA	LumB	HER2	Basal	LumA	LumB	HER2	Basal	LumA	LumB	HER2	Basal	LumA	LumB	HER2	Basal	LumA	LumB	HER2	Basal
LASSO-mCGA	98.92	97.85	100	100	98.95	97.99	100	100	98.92	97.89	100	100	0.9892	0.9785	1.00	1.00	0.9878	0.9731	1.00	1.00
PAM50	96.77	97.31	97.85	100	96.77	97.28	97.80	100	96.77	97.31	97.85	100	0.9677	0.9729	0.9779	1.00	0.9677	0.9493	0.9219	1.00

TABLE 4. Important KEGG pathways of the identified biomarkers.

Subtype	KEGG Pathway	Biomarkers
LumA	Chemical carcinogenesis	CYP2A6, PGR, CDC6
	Breast cancer	PGR
	Systemic lupus erythematosus	C4B, CTSG
	Bladder cancer, IL-17 signaling pathway, Pathways in cancer	MMP1
LumB	Chemical carcinogenesis	CYP2A6, BIRC5
	Colorectal cancer, Pathways in cancer, Hippo signaling pathway	BIRC5
	Transcriptional misregulation in cancer	H3C4
	Wnt signaling pathway	SFRP1
HER2	Breast cancer, Glioma, Prostate cancer, HIF-1 signaling pathway Proteoglycans in cancer, MAPK signaling pathway, PI3K-Akt signaling pathway Pathways in cancer, Melanoma	IGF1R
	PPAR signaling pathway	FABP7
Basal	Breast cancer, Gastric cancer, Hepatocellular carcinoma Human papillomavirus infection, Basal cell carcinoma	FZD9
	IL-17 signaling pathway	MUC5B, CXCL5, MAPK4
	Proteoglycans in cancer	ERBB4, FZD9
	Chemical carcinogenesis	CYP2A7
	Pathways in cancer	EPO, FZD9
	PI3K-Akt signaling pathway	EPO, ERBB4
	MAPK signaling pathway	ERBB4

while there are 2 common genes between HER2 and PAM50 (FOXC1, GRB7)

B. KEGG PATHWAY ANALYSIS

KEGG pathway analysis reveals the various disease pathways of the identified biomarkers for each subtype. Such analysis is performed using Enrichr tool.[2] Table 4 provides important KEGG pathways of the identified biomarkers for the four subtypes.

As is evident from Table 4, many of the predicted biomarkers for each subtype are not only enriched in the pathway for breast cancer but for other types of cancer as well. One of the identified biomarkers for LumA, CYP2A6, is known to be associated with lung cancer [36]. Many studies have attributed PGR gene in playing a significant role in breast cancer [37], [38], [39]. This is observed in the present study as well. PGR is also found to be differentially expressed in different cancer types like prostrate cancer [40] and lung cancer [41]. Deregulation of CDC6 has also a known role in posing serious risk of carcinogenesis [42]. CDC6 is a DNA initiation replication factor and when overexposed it shows oncogenic activities [43]. CDC6 disruption can also lead to chromosome instability in pancreatic cancer cells [44] and can result in glioma [45]. C4B and CTSG are enriched in the pathway of systemic lupus erythematosus; this is also confirmed by the findings in [46], [47], and [48].

[2]https://maayanlab.cloud/Enrichr/

TABLE 5. GO enrichment analysis for biological process, cellular component and molecular function of the identified biomarkers.

Subtype	Go Enrichment Analysis	Biomarkers
Biological Process		
LumA	Regulation of tumor necrosis factor superfamily cytokine production (GO:1903555)	CX3CR1, LTF
	Extracellular matrix disassembly (GO:0022617)	MMP1, CTSG, CTSV
	Regulation of glial cell migration (GO:1903975)	CX3CR1
	Regulation of microglial cell mediated cytotoxicity (GO:1904149)	CX3CR1
	Epoxygenase P450 pathway (GO:0019373)	CYP2A6
LumB	Regulation of tumor necrosis factor superfamily cytokine production (GO:1903555)	CX3CR1, LTF
	Positive regulation of transcription by RNA polymerase II (GO:0045944)	TFAP2B, FOXI1, FOXJ1, GRHL3, MYBL1, E2F7
	Positive regulation of fibroblast apoptotic process (GO:2000271)	SFRP1
	Negative regulation of apoptotic process (GO:0043066)	TFAP2B, SFRP1, BIRC5, CTSG
	Activation of GTPase activity (GO:0090630)	FOXJ1, CXCL13
	Positive regulation of non-canonical Wnt signaling pathway (GO:2000052)	SFRP1
HER2	Negative Regulation Of Mitotic Cell Cycle (GO:0045930)	FOXC1, BRINP3
	Transmembrane Receptor Protein Tyrosine Kinase Signaling Pathway (GO:0007169)	GRB7, FOXC1, IGF1R
	Negative Regulation Of Cell Cycle (GO:0045786)	FOXC1, BRINP3
	Regulation Of Mitotic Cell Cycle (GO:0007346)	FOXC1, BRINP3
Basal	Negative Regulation Of Myeloid Cell Apoptotic Process (GO:0033033)	EPO
	Negative Regulation Of MAP Kinase Activity (GO:0043407)	SERPINB3, BMP7
	Negative Regulation Of DNA-binding Transcription Factor Activity (GO:0043433)	FOXJ1, SIK1, BMP7
	Regulation Of Cellular Response To Oxidative Stress (GO:1900407)	BMP7
Cellular Component		
LumA	Endocytic Vesicle Lumen (GO:0071682)	SAA1, HBB, LTF
	Vacuolar Lumen (GO:0005775)	OGN, CTSG, CTSV
	Tertiary Granule Lumen (GO:1904724)	HBB, LTF
	Lysosomal Lumen (GO:0043202)	OGN, CTSV
	Secretory Granule Lumen (GO:0034774)	GRP, CTSG, LTF
LumB	Extracellular Vesicle (GO:1903561)	OGN, DEFB1
	Collagen-Containing Extracellular Matrix (GO:0062023)	SFRP1, SLPI, OGN, CTSG
	Golgi Lumen (GO:0005796)	OGN, DEFB1
HER2	Intermediate Filament (GO:0005882)	KRT, KRT20
	External Side Of Apical Plasma Membrane (GO:0098591)	LCT
	Protein Kinase Complex (GO:1902911)	IGF1R
	Transferase Complex, Transferring Phosphorus-Containing Groups (GO:0061695)	IGF1R
Basal	Basolateral Plasma Membrane (GO:0016323)	ERBB4, CA4, ENPP1, AQP3
	Brush Border Membrane (GO:0031526)	CA4, LRP2
	G Protein-Coupled Receptor Dimeric Complex (GO:0038037)	GABBR2
	Clathrin-Coated Endocytic Vesicle Membrane (GO:0030669)	LRP2, KIAA0319
Molecular Function		
LumA	Serine-Type Endopeptidase Activity (GO:0004252)	TPSD1, MMP1, CTSG, CTSV
	Cysteine-Type Endopeptidase Inhibitor Activity (GO:0004869)	CST2, LTF
	Endopeptidase Activity (GO:0004175)	TPSD1, MMP1, CTSG, CTSV
	Alcohol Dehydrogenase Activity, Zinc-Dependent (GO:0004024)	ADH1B
LumB	DNA-binding Transcription Activator Activity, RNA Polymerase II-specific (GO:0001228)	TFAP2B, FOXI1, FOXJ1, GRHL3, MYBL1
	CCR Chemokine Receptor Binding (GO:0048020)	DEFB1, CXCL13
	CCR6 Chemokine Receptor Binding (GO:0031731)	DEFB1
	Transcription Cis-Regulatory Region Binding (GO:0000976)	TFAP2B, FOXI1, FOXJ1, E2F7
HER2	Gap Junction Channel Activity Involved In Cell Communication By Electrical Coupling (GO:1903763)	GJB6
	Aldo-Keto Reductase (NADP) Activity (GO:0004033)	AKR1B10
	Hexose Transmembrane Transporter Activity (GO:0015149)	SLC5A1
	Alcohol Dehydrogenase (NADP+) Activity (GO:0008106)	AKR1B10
Basal	Neuropeptide Receptor Binding (GO:0071855)	PRLH, NTS
	Thiamine Transmembrane Transporter Activity (GO:0015234)	SLC44A4
	ABC-type Glutathione S-conjugate Transporter Activity (GO:0015431)	ABCC11
	cAMP Response Element Binding Protein Binding (GO:0008140)	SIK1

MMP1 is unique biomarker in bladder cancer as shown by Jin et al. in [49] and Wallard et al. in [50]. For LumB subtype, BIRC5 which is identified as one of the potential biomarkers is a well-known therapeutic target for cancer [51] which is also shown to be associated with tumour immune cell infiltration [52]. BIRC5 is enriched in the pathway of colorectal cancer; this is also corroborated by the findings in [53] and [54]. Another biomarker SFRP1 to identify LumB is enriched in Wnt signaling pathway as also reported in [55], [56], and [57]. Apart from breast cancer, IGF1R which is an identified biomarker for HER2, is also enriched in pathways for MAPK signaling, PI3K-Akt signaling, pathways in cancer and melanoma. Such findings have also been reported in [58], [59], [60], and [61]. On the other hand, another biomarker FABP7 is enriched in PPAR signalling pathway [62]. FZD9, a biomarker for Basal subtype is enriched in pathways for breast cancer, gastric cancer, hepatocellular carcinoma, human papillomavirus infection and basal cell carcinoma. The results of Kirikoshi et al. [63] show that FZD9 might play key roles in human gastric cancer while [64] has reported the involvement of FZD9 in hepatocellular carcinoma. László et al. [65] have reported

that FZD9 becomes upregulated in human papillomavirus infection. ERBB4 is enriched in the pathways of PI3K-Akt signaling and MAPK signaling. Xu et al. [66] has reported that high levels of ERBB4 were noticed in gastric cancer patients from which they concluded that ERBB4 promotes the proliferation of such cancer cells through PI3K-Akt signaling pathway. Forget et al. [67] have also reported the role of ERBB4 in MAPK and PI3K-Akt signaling pathways.

C. GO ENRICHMENT ANALYSIS

Table 5 reports the GO Enrichment analysis of the identified biomarkers for the four subtypes. Like KEGG, this analysis is also performed using Enrichr tool. GO analysis helps to understand the biological, cellular and molecular processes associated with the identified biomarkers.

Some significant GO terms related to biological process for the subtypes for example, LumA are regulation of extracellular matrix disassembly (GO:0022617), regulation of glial cell migration (GO:1903975), regulation of microglial cell mediated cytotoxicity (GO:1904149) and epoxygenase P450 pathway (GO:0019373). These GO terms are mostly related to breast cancer. Some significant cellular components are endocytic vesicle lumen (GO:0071682), vacuolar lumen (GO:0005775), tertiary granule lumen (GO:1904724), lysosomal lumen (GO:0043202), secretory granule lumen (GO:0034774) while molecular functions include serine-type endopeptidase activity (GO:0004252), cysteine-type endopeptidase inhibitor activity (GO:0004869), endopeptidase activity (GO:0004175) and Alcohol Dehydrogenase Activity, Zinc-Dependent (GO:0004024). Although, GO enrichment analysis is a powerful tool for understanding the significance of genes, it has certain limitations where multiple hypothesis testing can lead to false positives that can affect the interpretation of results. In such cases, statistical tests like Benjamini-Hochberg false discovery rate (FDR) or Bonferroni correction can help to provide better interpretation.

IV. CONCLUSION

Breast Cancer is the leading cause of cancer related cases worldwide. Thus, cancer prediction in its early stages may help in halting the progression of the disease. In this work, we have proposed a LASSO-based modified compact genetic algorithm (LASSo-mCGA) to identify the set of biomarkers for each breast cancer subtype. Initially, by applying LASSO on 28201 genes, reduced set of genes, 97 for LumA, 80 for LumB, 126 for HER2 and 188 for BASAL are identified. Subsequently, four machine learning algorithms, RF, SVM, KNN and NB are applied on such reduced sets to identify the best machine learning algorithm. With accuracy values of 95.63% for LumA, 94.85% for LumB, 98.77% for HER2 and 99.76% for Basal, SVM is clear winner. Thereafter, the number of final set of biomarkers as identified by mCGA and SVM are 50, 45, 33 and 98 for LumA, LumB, HER2 and Basal respectively. The final classification accuracy values are 98.92% for LumA, 97.85% for LumB and 100% for both HER2 and Basal. SVM is also applied on the genes as identified by PAM50 method and the results clearly show that a common set of genes is unable to identify breast cancer subtype. Finally, KEGG pathway analysis and GO enrichment analysis are carried out to validate the biological significance of the identified biomarkers. As a future research work, we intend to extend our work for larger dataset as TCGA dataset is relatively small. We also intend to apply the proposed methodology on other cancer subtypes. Furthermore, we also would like to apply the proposed methodology for genetic and epigenetic biomarkers which can significantly enhance our ability to diagnose, prognose and treat diseases effectively.

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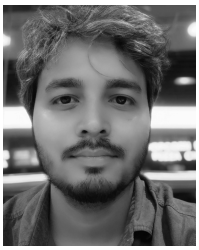
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